

Sígrid Vila Bagaria

PhD Candidate in Medical Image Analysis & Computational Pathology
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Computer vision engineer developing AI pipelines for oncologic image analysis, from 3D generative MRI synthesis to whole-slide histopathology. My current work links H&E slides with transcriptomic immunotherapy signatures through cross-modal learning, and tackles tumour subtyping, biomarker (HER2) assessment and cell-level analysis with clinical partners.

RESEARCH INTERESTS

Computational pathology · multimodal learning (histology–omics) · biomarker discovery from WSIs · multiple instance learning · generative models (GANs, diffusion) · cell detection & classification

PUBLICATIONS

2026 S. Vila-Bagaria, M. Teixidó, M. Piñol, F. Vilardell, R. Montal, V. Vilaplana. *Cross-Modal Contrastive Learning for the Retrieval of Immunotherapy-Associated Molecular Signatures from Histopathology. MICCAI 2026 — CAPTION Workshop.*

2026 S. Vila-Bagaria, M. Teixidó, M. Piñol, F. Vilardell, R. Montal, V. Vilaplana. *Zero-Cost Virtual RNA: Approximating Immunotherapy Signatures via Cross-Modal WSI Retrieval. MIDL 2026 — Short Papers.*

2026 M. Ysern*, S. Vila-Bagaria*, O. Pina, V. Vilaplana. *Dual-CellNucDETR: Context-Aware Object-Level Cell Analysis. MICCAI 2026 — COMPAYL Workshop.* (*equal contribution)

2025 C. Giardina, S. Vila-Bagaria, C. Albornés, M. Pardàs, V. Vilaplana. *Anatomically Guided 3D Diffusion Models for Synthetic Prostate MRI Generation. MICCAI 2025 — Workshop on Deep Generative Models, pp. 288–297.* doi:10.1007/978-3-032-05472-2_28

Theses S. Vila Bagaria. *Synthesis of Prostate MRI with Generative Adversarial Networks. MSc Thesis, 2024 — ACMCV Best Thesis Award.*

S. Vila Bagaria. *Visual Sport Analysis Using Computer Vision and Deep Learning Techniques. BSc Thesis, UPCommons, 2023.*

EDUCATION

2025 – Present

PhD in Medical Imaging & Computer Vision

Universitat Politècnica de Catalunya (UPC), Image Processing Group (GPI). Supervisor: Verónica Vilaplana.

- AI pipelines for H&E histopathology classification and subtyping, cell detection and classification, and MRI generation.
- Funded by the competitive **FI-STEP** PhD grant (Generalitat de Catalunya).

2023 – 2024

MSc in Computer Vision

Computer Vision Center (CVC), Universitat Autònoma de Barcelona

- 9.3/10 — **Best Student Record Award**; Distinction (Matrícula d'Honor) in Introduction to Human & Computer Vision, 3D Vision, Visual Recognition, Video Analysis and the Master's Thesis.
- Thesis: *Synthesis of Prostate MRI with Generative Adversarial Networks* — **Best Master Thesis Award** (ACMCV).

2019 – 2023

BSc in Data Science and Engineering

Universitat Politècnica de Catalunya (UPC)

- Algorithms, signals & systems, optimisation, machine/deep learning, computer vision, NLP, databases. 7.77/10.
- Erasmus+ exchange at Chalmers University of Technology, Sweden (2022).
- Thesis: *Visual Sport Analysis Using Computer Vision and Deep Learning Techniques*.

RESEARCH EXPERIENCE

2025 – Present

PhD Researcher

UPC, Image Processing Group (GPI), Barcelona

- **Gastric adenocarcinoma subtyping & virtual RNA** (with IRBLleida): MIL pipeline that classifies H&E whole-slide images into genetic subtypes, and cross-modal contrastive learning to retrieve immunotherapy-associated transcriptomic signatures directly from histology (MICCAI 2026 workshop, MIDL 2026).
- **Spatial analysis of HER2 expression**: WSI pipeline to analyse and score HER2 status and its spatial patterns in endometrial tissue, aimed at a clinical application.
- **Endometrial cell classification** (DigiPatICS project, Institut Català de la Salut): context-aware DETR-based cell and nucleus detection and classification to support diagnosis (MICCAI 2026 workshop).

2024 – 2025

Research Engineer

UPC, Image Processing Group (GPI), Barcelona

- **Prostate MRI generation** within the EU Horizon project **FLUTE** (federated learning and multi-party computation for prostate cancer diagnostics): GAN and anatomically guided 3D diffusion models for synthetic MRI, latent-space control, and data augmentation for downstream classification (MICCAI 2025 workshop).

Mar 2023 – Nov 2023

Research Assistant (INIREC Scholarship)

UPC, Image Processing Group (GPI), Barcelona

- Online and offline player and ball tracking for basketball broadcasts in collaboration with MediaPro; improved SORT-based tracking to reduce re-identification errors.

AWARDS & GRANTS

2025 **FI-STEP PhD Grant** — Generalitat de Catalunya, competitive doctoral funding.

2024 **Best Master Thesis Award** — Annual Catalan Meeting in Computer Vision (ACMCV).
Best Student Record Award — MSc in Computer Vision, CVC.

2023 **INIREC Research Scholarship** — UPC.

2022 **Erasmus+ Exchange** — Chalmers University of Technology, Sweden.

TEACHING

2026 – Present **Machine Learning I** — BSc Data Science and Engineering, UPC. Lab sessions, mentoring, assessment.

2025 – Present **Deep Learning for AI** — MSc Telecommunications Engineering & MSc ATT, UPC. Lab sessions, mentoring, assessment.
Deep Learning & Computer Vision — Postgraduate in AI with Deep Learning, UPC School. Lab sessions; advisor of a final-project group.

SKILLS

Programming — Python (PyTorch, NumPy/SciPy, OpenCV, scikit-image), Git, Slurm/HPC, MATLAB, R, C++, SQL

Methods — Whole-slide image analysis, multiple instance learning, contrastive/cross-modal learning, detection transformers, GANs and diffusion models, 3D MRI, object tracking, statistical analysis

Languages — Catalan (native) · Spanish (native) · English (C1) · French (B2)